

ES 192: Mini-Project Brief

The Multi-Material Skateboard Deck Challenge

Materials Selection in Mechanical Design

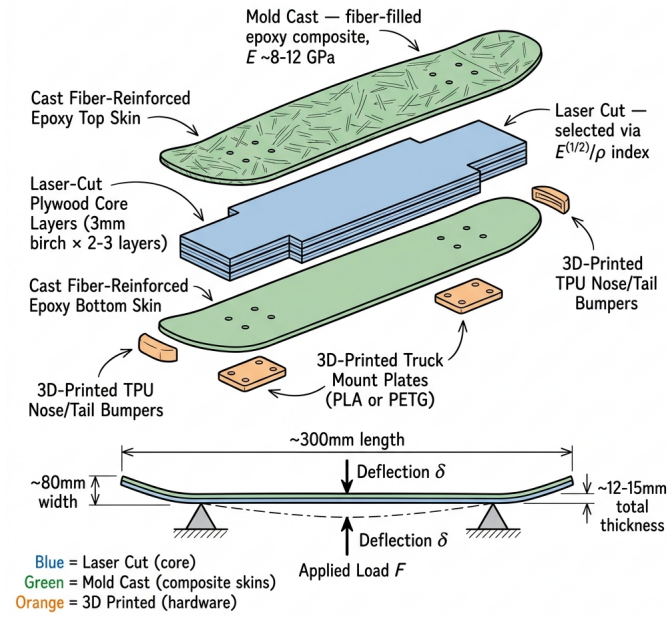
Harvard SEAS - Fall 2026

Prepared by: Seymour Hasanov

For discussion with: Nora Cullen, Director for Active Learning

Draft - April 2026

Mini Skateboard Deck — Multi-Material Concept



1. Concept Overview

Each team of 3 students designs, fabricates, and tests a multi-material mini skateboard deck that must maximize load-carrying capacity while minimizing weight. The deck is a sandwich panel: laser-cut core layers bonded to cast composite skins, with 3D-printed mounting hardware.

What makes this a materials selection project (not just a build project): every component's material is chosen using the Ashby methodology - students derive materials indices, screen candidates on property charts, and justify their choices with measured data. They then validate predictions against physical test results.

Why a Skateboard Deck?

- ? It is fundamentally a beam/panel under bending - the same mechanics as a cantilever, but far more engaging for students
- ? The sandwich structure (skins + core) directly teaches composite and hybrid material concepts (Ch 13-14)
- ? 3-point bend testing is standard and requires a simpler fixture than a cantilever clamp
- ? Students can hold it, flex it, and keep it - tangible and rewarding
- ? Directly leads into the "Composite longboard deck" group project option listed in the syllabus

Performance Metric

$$P = F_{\text{max}} / m$$

- ? F_{max} = maximum load supported without exceeding 5 mm center deflection
- ? m = total mass of the deck assembly
- ? Secondary metric: load at structural failure / mass

2. Specifications

Parameter	Value
Deck length	~300 mm (mini/cruiser scale)
Deck width	~80 mm
Total thickness	~12-15 mm (core + skins)
Shape	Flat (no concave, no kick tails)
Test method	3-point bend: supports at truck positions (~200 mm span), load at center
Max allowable deflection	5 mm under service load
Performance metric	$P = F_{\text{max}} / m$
Manufacturing processes	>= 3 required: laser cutting, 3D printing, silicone mold casting
Materials	Standard ALL stock provided (see Section 4)

3. Structural Concept - Exploded View

The deck is a sandwich panel - the same architecture used in real performance skateboard decks, aerospace panels, and composite structures. Each layer serves a specific structural function and is made by a different manufacturing process.

Mini Skateboard Deck — Multi-Material Concept

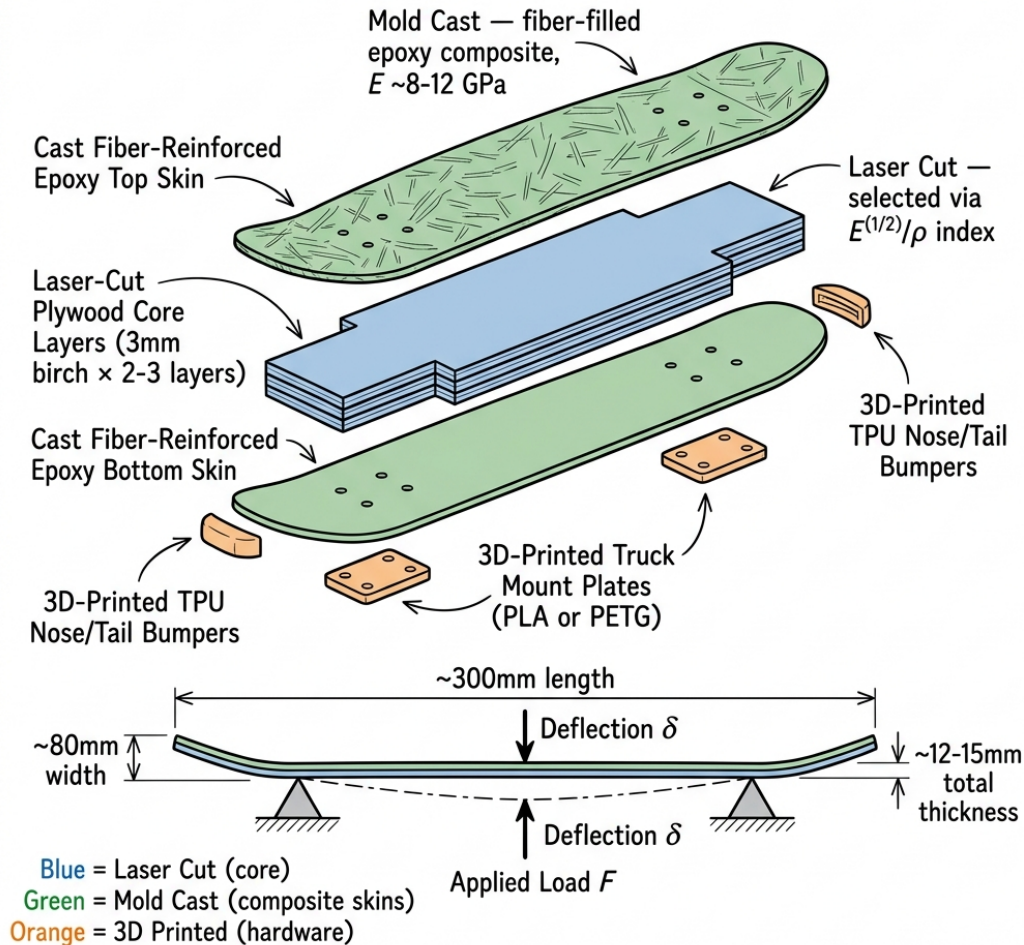


Figure 1: Exploded isometric view of the multi-material skateboard deck showing laser-cut plywood core (blue), cast fiber-reinforced epoxy skins (green), and 3D-printed hardware (orange). Bottom: assembled side profile with 3-point bend test setup.

Sandwich Panel Mechanics

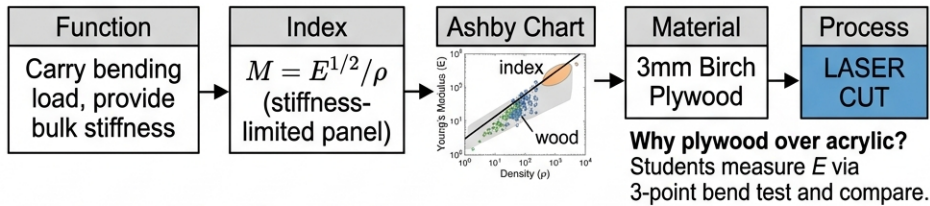
- ? Top & bottom skins (cast composite): carry tension and compression from bending
- ? Core layers (laser-cut plywood): carry shear, provide bulk thickness to separate skins
- ? Hardware (3D-printed): truck mounts transfer load, bumpers absorb impact

This sandwich architecture is a direct application of the course's hybrid materials content (Chapters 13-14). Students learn that separating stiff skins with a lightweight core dramatically increases bending stiffness with minimal weight penalty - a concept they can feel by flexing their deck.

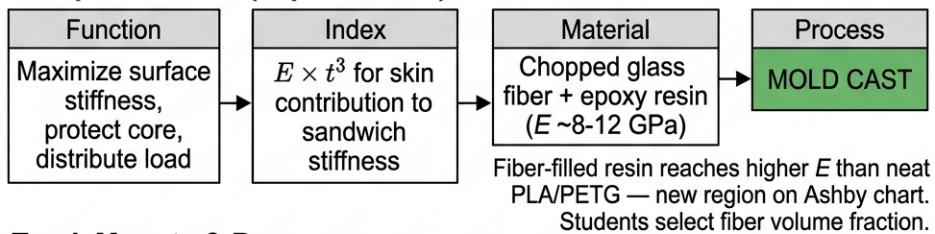
4. Materials Selection Logic

The Ashby methodology drives the material-to-process assignment for each component. Each component has a function, which translates to a materials index, which points to candidate materials, which constrain the manufacturing process. The diagram below shows this chain for all three component groups.

Core Layers



Composite Skins (Top + Bottom)



Truck Mounts & Bumpers

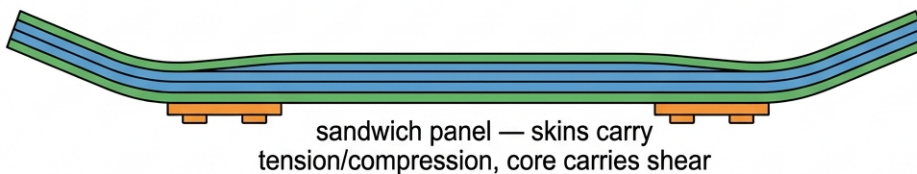
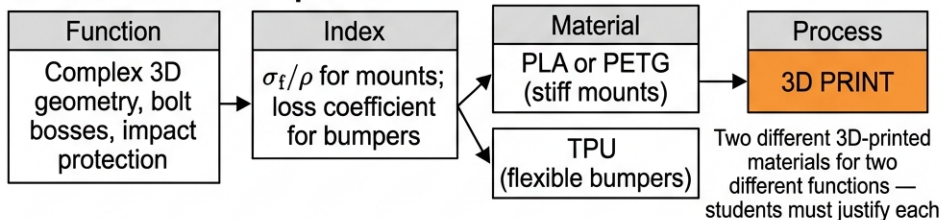


Figure 2: Materials selection logic - Function -> Index -> Ashby Chart -> Material -> Process for each component. Bottom: cross-section showing sandwich panel mechanics.

5. Material Options - The Decision Space

The whole point of the Ashby exercise is that students have multiple real candidates to choose from and must justify their selection with data. Below is the full palette of materials available in the ALL, organized by manufacturing process.

SECTION 1 — “CORE LAYERS (Laser Cut)”

Birch Plywood 3mm

— E: 12 GPa
 ρ : 600 kg/m³
 — good E/ ρ , natural fiber composite

Acrylic (PMMA) 3mm

— E: 3.2 GPa
 ρ : 1180 kg/m³
 — transparent, brittle

MDF 3mm

— E: 4 GPa
 ρ : 750 kg/m³
 — uniform, no grain direction

Aluminum 1.5mm

— E: 69 GPa
 ρ : 2700 kg/m³
 — highest E, heaviest

? Which has the best $E^{1/2}/\rho$? Students must plot on Ashby chart and **JUSTIFY**

SECTION 2 — “COMPOSITE SKINS (Mold Cast)”

Neat Epoxy

— E: ~3 GPa
 baseline, no filler

Rigid Polyurethane

— E: ~2.5 GPa
 fast cure, lower cost

Chopped Glass + Epoxy

— E: ~8-12 GPa
 short fiber composite, highest stiffness

Flexible Polyurethane

— E: ~0.01 GPa
 grip/damping layer, NOT structural

Mineral-Filled Epoxy

— E: ~5-6 GPa
 stiffer than neat, cheaper than glass

? Structural skin or damping layer?
 Different functions → different material indices → different choices

SECTION 3 — “HARDWARE (3D Printed)”

PLA

— E: 3.5 GPa
 σ : 60 MPa
 stiffest, easiest to print, brittle

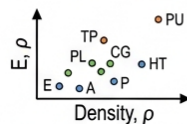
PETG

— E: 2.2 GPa
 σ : 50 MPa
 tougher, better layer adhesion, slight flex

TPU

— E: 0.03 GPa
 σ : 30 MPa
 flexible, impact absorbing, hard to print

? Truck mounts need stiffness (PLA or PETG?). Bumpers need flexibility (TPU).
 Two different functions → two different materials from same process



Sketch approximate positions of **ALL** the mentioned materials labeled in windir candidate pool span: a **wide range**, students must employ index lines for precise selection

Figure 3: Complete material candidate pool organized by manufacturing process, with key properties and selection questions for each category.

5.1 Core Layers - Laser Cut (students pick 1-2)

Material	E (GPa)	Density (kg/m3)	$E^{1/2}/\rho$	Notes
Birch plywood (3mm)	12	600	5.8×10^{-3}	Natural fiber composite, best index
Acrylic / PMMA (3mm)	3.2	1180	1.5×10^{-3}	Transparent, brittle, poor index
MDF (3mm)	4	750	2.7×10^{-3}	Uniform (no grain), heavy
Aluminum (1.5mm)	69	2700	3.1×10^{-3}	Highest E, but heaviest

Selection question: Plywood wins on $E^{1/2}/\rho$, but aluminum has 6x higher absolute E. When does raw stiffness matter more than specific stiffness? Students test coupons in Week 2 to decide.

5.2 Composite Skins - Mold Cast (students pick 1 per skin)

Material	E (GPa)	Density (kg/m3)	Notes
Neat epoxy	~3	~1200	Baseline, no filler, easy to pour
Rigid polyurethane	~2.5	~1100	Fast cure (~30 min), lower cost
Chopped glass + epoxy	~8-12	~1600	Short fiber composite, highest stiffness
Mineral-filled epoxy	~5-6	~1500	Stiffer than neat, cheaper than glass
Flexible polyurethane	~0.01	~1050	NOT structural - grip/damping layer

Selection question: Max stiffness (glass-filled bottom skin) or trade top-skin stiffness for a flexible grip layer (flexible PU on top)? Different functions -> different indices -> different choices.

5.3 Hardware - 3D Printed (students pick per component)

Material	E (GPa)	Strength (MPa)	Notes
PLA	3.5	60	Stiffest, easiest to print, brittle - cracks on impact
PETG	2.2	50	Tougher, better layer adhesion, slight flex
TPU	0.03	30	Flexible, impact absorbing - bumpers only?

Selection question: Truck mounts see bolt preload + vibration. PLA is stiffer but PETG won't crack. Students justify using σ_f/ρ AND consider failure mode. Bumpers are a separate function - loss coefficient matters, not E.

6. Lab Schedule Overview (6 Labs)

The mini-project runs over 6 weekly labs (2 hours each, ~10 students / 3-4 teams per section). Each lab has two parallel tracks: a manufacturing skill and a materials selection activity. The selection thread builds cumulatively: Screen -> Measure -> Validate -> Select -> Compare -> Simulate.

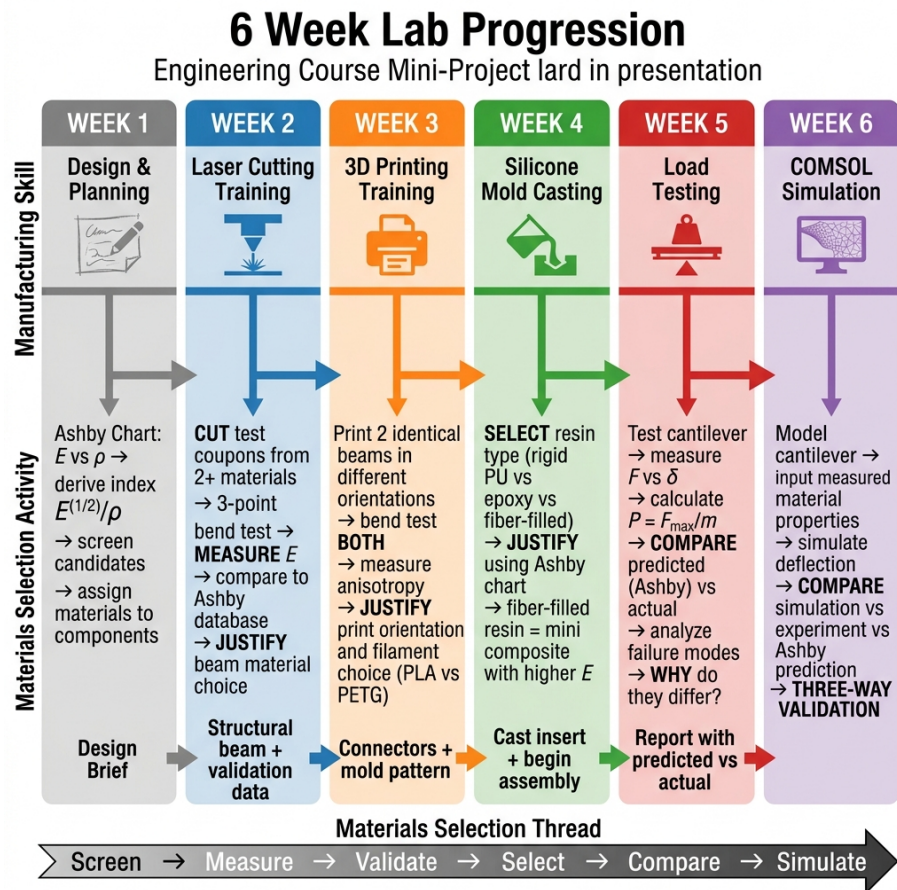


Figure 4: Six-week lab progression showing manufacturing skills (top track) and materials selection activities (bottom track) converging at each week.

7. Detailed Lab Plans

Lab 1 - Design & Planning (Week 1)

Objectives

- ? Students are safe to work in the ALL and understand all equipment
- ? Teams form and understand the skateboard deck challenge
- ? Each team produces a design plan with Ashby-based material justification

Selection Concept: SCREEN

Students perform functional decomposition of the deck (core, skins, hardware), fill out a Design Requirements Table (Function / Constraints / Objectives / Free Variables) for each component, plot an E vs. ρ Ashby chart, derive the beam index $M = E^{1/2}/\rho$, overlay it on the chart, and screen candidates from the available stock materials.

Plan (2 hours)

Time	Activity	Details
0:00-0:30	ALL Tour + Safety	Walk through laser bay, 3D print lab, casting station, workshop. Safety rules. Sign agreements.
0:30-0:45	Project Briefing	Present the Skateboard Deck Challenge: specs, performance metric, test setup, timeline, grading.
0:45-1:00	Team Formation	Form teams of 3. Concept sketching: what shape? How many core layers? Where do skins go?
1:00-1:30	Design + Selection	Design Requirements Tables per component. Ashby E vs ρ chart. Derive $E^{1/2}/\rho$ index. Screen candidates.
1:30-1:50	Design Decisions	Assign material + process to each component. Begin rough dimensioning. CAs challenge choices.
1:50-2:00	Wrap-up	Deliverable: Design Brief (1-2 pages) due before next lab.

Materials & Equipment Needed

- ? Whiteboards or large paper, markers
- ? Printed material property data sheets (E, ρ , σ_f for all stock)
- ? Rulers, example deck photos (good and bad)
- ? Materials selection software access (if available)

Deliverable

Design Brief (1-2 pages): annotated sketch, Design Requirements Table per component, Ashby chart with index line and candidates highlighted, material/process justification.

Lab 2 - Laser Cutting: Core Layers (Week 2)

Objectives

- ? Students can safely operate the laser cutter
- ? Core layers are cut and test coupons prepared
- ? Students have measured material properties to validate their Ashby chart predictions

Selection Concept: MEASURE

Students cut test coupons (100 x 20 mm) from at least 2 different sheet materials, perform 3-point bend tests to measure flexural modulus E, and compare measured values to the Ashby database. This forces them to ask: "Is the E value in the chart accurate for this specific stock, cut this way?"

Plan (2 hours)

Time	Activity	Details
0:00-0:20	Laser Cutter Training	Demo by CA: operation, material loading, focus, DXF/SVG import, kerf compensation, speed/power per material. Safety rules.
0:20-0:40	File Preparation	Teams prep 2D cut files: core layer profiles (could vary width along length) + test coupons (100x20mm from 2+ materials).
0:40-1:10	Cutting	Teams take turns cutting. Available: 3mm plywood, 3mm acrylic, 3mm MDF, 1.5mm aluminum.
1:10-1:40	Baseline Testing	3-point bend test on coupons: 80mm span, load at center, measure deflection at 3+ load increments. Calculate $E = FL^3/(48\delta I)$.
1:40-1:55	Discussion	Which material had best specific stiffness? Did measurements match database? How does laser cutting constrain geometry?
1:55-2:00	Wrap-up	Verify all teams have cut core layers + test data. Remind: bring CAD files for printing next week.

Materials & Equipment Needed

- ? Sheet stock: 3mm birch plywood, 3mm acrylic, 3mm MDF, 1.5mm aluminum
- ? 3-point bend test blocks (2 support blocks + loading nose)
- ? Calibrated weights (100g, 200g, 500g, 1kg)
- ? Dial indicator or rulers, digital scales, calipers

Deliverable

Laser-cut core layers + measured flexural modulus data for 2+ materials (data table + comparison to Ashby database).

Lab 3 - 3D Printing: Hardware & Mold Pattern (Week 3)

Objectives

- ? Students can use FDM printers and slicer software
- ? Truck mounts, bumpers, and mold pattern are printing
- ? Students understand how print orientation changes material properties (anisotropy)

Selection Concept: VALIDATE PROCESS-PROPERTY EFFECTS

Students print two identical test beams in PLA - one flat (layers parallel to length) and one on-edge (layers perpendicular). They bend-test both and measure the stiffness difference. This demonstrates that the same material has different effective properties depending on how it's processed - the Ashby chart gives bulk values, but process changes them.

Plan (2 hours)

Time	Activity	Details
0:00-0:20	3D Printing Training	Demo: slicer walkthrough (STL import, layer height, infill, orientation, supports, brim). Filaments: PLA, PETG, TPU - when to use each.
0:20-0:35	Design Review + Slicing	CAs review designs for printability. Each team preps: (A) truck mounts/bumpers, (B) mold pattern for skin casting (with draft angles for demolding).
0:35-0:55	Orientation Experiment	Print 2 small test beams (60x10x5mm) in PLA: flat vs. on-edge (~10 min each). Discuss anisotropy while printing.
0:55-1:15	Test Orientation Beams	3-point bend test both beams. Calculate stiffness ratio. The flat beam will be noticeably stiffer. Record data.
1:15-1:45	Start Functional Prints	Queue truck mounts, bumpers, mold pattern. Small parts may finish; larger prints run after session.
1:45-2:00	Wrap-up	Confirm mold patterns will be ready before Week 4. Preview casting process. Bring ALL components to Week 4.

Materials & Equipment Needed

- ? PLA, PETG, TPU filament spools
- ? FDM printers (2-3 per section)
- ? Slicer software on lab computers
- ? Pre-made test beam STL file (shared via Canvas)
- ? 3-point bend test setup from Week 2

Deliverable

3D-printed truck mounts + bumpers + mold pattern + print orientation test data (stiffness anisotropy ratio).

Lab 4 - Silicone Mold Casting & Assembly (Week 4)

Objectives

- ? Students can make silicone molds and cast resin parts
- ? Composite skin layers are cast (material choice justified via Ashby)
- ? Deck assembly begins - bonding skins to core

Selection Concept: CREATE NEW MATERIALS

Students select a resin system for their skins - a genuine Ashby exercise. They can create a fiber-reinforced composite (chopped glass + epoxy, $E \sim 8\text{-}12$ GPa) that occupies a different region on the Ashby chart than any neat FDM filament. They're not just picking from a list - they're designing a material by choosing fiber type and volume fraction.

Plan (2 hours)

Time	Activity	Details
0:00-0:15	Casting Training	Demo: mold release on pattern, containment box (foam core + hot glue), mix silicone by weight, pour, degas if possible. Cure: 4-6 hrs.
0:15-0:40	Mold-Making	Teams pour silicone molds for flat skin panels. Teams with already-cured molds proceed to resin casting. Backup molds available.
0:40-1:10	Resin Casting	Select resin: neat epoxy, rigid PU, fiber-filled epoxy, mineral-filled, or flexible PU. Mix (add fibers if chosen). Pour into mold. Cure 30-60 min initial.
1:10-1:40	Begin Assembly	While skins cure: dry-fit core layers + truck mounts. Plan bonding strategy (epoxy adhesive, bolts, press-fit). Begin laminating core layers.
1:40-1:55	Discussion	What shapes does casting enable that other processes cannot? Defects? How does resin choice affect performance?
1:55-2:00	Wrap-up	CAs will demold cured parts. Teams finalize assembly before Week 5. Open lab time may be arranged.

Materials & Equipment Needed

- ? Silicone rubber (two-part, e.g., Smooth-On Mold Star)
- ? Mold release spray, foam core + hot glue guns (containment)
- ? Casting resins: rigid PU, flexible PU, epoxy, chopped glass fiber, mineral filler
- ? Mixing cups, stir sticks, digital scale, disposable gloves, eye protection
- ? Vacuum degassing chamber (if available)
- ? Pre-made backup silicone molds (2-3 per section)

Deliverable

Cast composite skin layers + resin selection justification on Ashby chart + partially assembled deck.

Lab 5 - Load Testing & Validation (Week 5)

Objectives

- ? Every team has a fully assembled, tested deck with measured data
- ? Students compare Ashby predictions to actual performance
- ? Students analyze failure modes and reflect on methodology

Selection Concept: COMPARE PREDICTED vs. ACTUAL

This is the core learning moment. Students compare three things: (1) their Week 1 Ashby-based prediction of performance, (2) their actual test result, and (3) why they differ. Common reasons: joints fail before materials, cast parts have voids, print layers delaminate, bonding is imperfect. They learn where the methodology works and where real-world factors dominate.

Plan (2 hours)

Time	Activity	Details
0:00-0:15	Final Assembly	Attach cured skins to core. Tighten truck mounts. Trim flash. Weigh completed deck. Measure dimensions.
0:15-0:20	Test Procedure Review	Support at truck positions (~200mm span). Load at center incrementally. Record F vs delta. Find F_max at delta=5mm.
0:20-1:00	Testing (all teams)	~10 min per team. While waiting: calculate performance index, review Week 1 predictions. If safe, load to failure.
1:00-1:20	Leaderboard	All teams calculate $P = F_{\text{max}}/m$. Post results. Rank by performance index. Celebrate top design.
1:20-1:50	Validation Discussion	Each team: Ashby prediction vs. actual result. Why different? Which choices mattered most? What would you change?
1:50-2:00	Report Assignment	Mini-project report (3-5 pages): design, Ashby plots, test data, predicted vs actual, failure modes, lessons.

Materials & Equipment Needed

- ? 3-point bend test fixture (2 support blocks at ~200mm span)
- ? Calibrated weights (100g through 5kg)
- ? Dial indicator + magnetic stand
- ? Digital scale, calipers
- ? Whiteboard for leaderboard
- ? Camera (students photograph failure modes for reports)

Deliverable

Mini-project report (3-5 pages, due in 1 week): design rationale, Ashby plots, material/process justification, load-deflection curve, predicted vs. actual analysis, failure mode discussion, lessons learned.

Lab 6 - COMSOL Simulation (Week 6)

Objectives

- ? Students can set up a basic structural simulation in COMSOL
- ? Students compare simulation to experiment and Ashby prediction (three-way validation)
- ? Students understand when simulation is useful vs. when physical testing is necessary

Selection Concept: SIMULATE & THREE-WAY VALIDATION

Students model their deck in COMSOL using their measured material properties (not textbook values), simulate the 3-point bend test, and compare the result to both their Ashby prediction and their experimental data. They see that Ashby and COMSOL roughly agree (both assume perfect material), but both differ from experiment (which includes joints, defects, imperfect bonding). This teaches when simulation adds value and when it doesn't replace testing.

Plan (2 hours) - Run by COMSOL Instruction Team

Time	Activity	Details
0:00-0:20	COMSOL Intro	Interface overview. Open pre-built sandwich panel template. Walk through: geometry, material assignment, BCs, load.
0:20-0:50	Guided Exercise	Students modify template: adjust deck dimensions, assign materials (E, nu, rho from their measured data), apply their test loads. Run.
0:50-1:10	Comparison Exercise	Compare COMSOL deflection to experimental data from Week 5. Fill comparison table: predicted vs measured at each load.
1:10-1:40	Parametric Study	Vary one parameter (core material, skin thickness, fiber volume fraction). Plot results. "What if I used aluminum core instead?"
1:40-2:00	Wrap-up Discussion	When is simulation useful vs. when is testing necessary? Preview COMSOL features for group projects.

Three-Way Comparison Table (students fill in)

Prediction Method	Source	Deflection at Load X
Ashby index (Week 1)	Analytical, from materials index	delta_predicted
COMSOL simulation (Week 6)	Computational, idealized geometry	delta_simulated
Experiment (Week 5)	Physical test, real structure	delta_measured

Materials & Equipment Needed

- ? COMSOL via HUIT VDI (student access set up in advance)
- ? Pre-built sandwich panel template (.mph file)
- ? Material property table matching lab stock
- ? Worksheet with step-by-step instructions + comparison table template

8. Materials Selection Thread Summary

The materials selection methodology is reinforced every single week. Each lab has a "selection verb" that builds on the previous week. By Week 5, students have gone through the complete Ashby cycle with physical evidence (test data + a broken deck) to prove it.

Week	Lab	Selection Verb	What Students Learn
1	Design	SCREEN	Use Ashby charts + indices to narrow candidates
2	Laser Cut	MEASURE	Real properties != database values; validate assumptions
3	3D Print	VALIDATE	Process changes properties (anisotropy); selection must account for mfg
4	Casting	CREATE	Design new materials (fiber composites) to access new chart regions
5	Testing	COMPARE	Prediction vs reality; where methodology works and breaks down
6	COMSOL	SIMULATE	Computational validation; parametric "what-if" exploration

9. Grading Rubric (10% of Course Grade)

Criterion	Weight	Description
Materials Selection Rationale	30%	Ashby charts, materials indices, Design Requirements Tables, justification quality
Design Quality & Manufacturing	25%	Creativity, execution quality, appropriate use of each process
Test Results & Performance	20%	Data quality, $P = F_{\max}/m$, rigor of testing protocol
Report & Analysis	25%	Clarity, predicted-vs-actual comparison, failure mode analysis, reflection

10. Pre-Semester Action Items

Task	Owner	Target
Finalize deck specs and test fixture design	Seymur	End of April 2026
Test-run full 6-week sequence; write lab handouts	Nora	End of May 2026
Build or procure 3-point bend test fixture	Nora + ALL staff	End of May 2026
Prepare 2-3 backup silicone molds (flat skin panel shape)	Nora	End of May 2026
Coordinate with COMSOL team: provide sandwich panel template specs	Seymur	End of April 2026
COMSOL team: build sandwich panel template .mph file	COMSOL team	End of May 2026
Recruit and train lab CAs from ES 51	Seymur + Nora	August 2026
Procure silicone, resins, chopped fiber, casting supplies	Nora + ALL staff	August 2026
Prepare material data sheets for students	Seymur	Before Week 1